

Lab 2

Configuration BGP entre trois systèmes autonomes

Cisco Packet Tracer - Protocole BGP

Routeurs	nw_R1, nw_R2, nw_R3 (Cisco 2811)
Systèmes autonomes	AS 100 (R1) / AS 200 (R2) / AS 300 (R3)
Protocole	eBGP : R1 - R2 et R2 - R3
Ordre du document	R1, R2, R3, puis tests depuis les PC

1. Objectif et topologie

Configurer eBGP entre trois systèmes autonomes alignés : R1 (AS 100), R2 (AS 200) et R3 (AS 300). Chaque routeur annonce sa loopback et son réseau local ; R2, au centre, relaie les routes entre R1 et R3. On valide ensuite avec les tables de routage et des pings entre les PC.

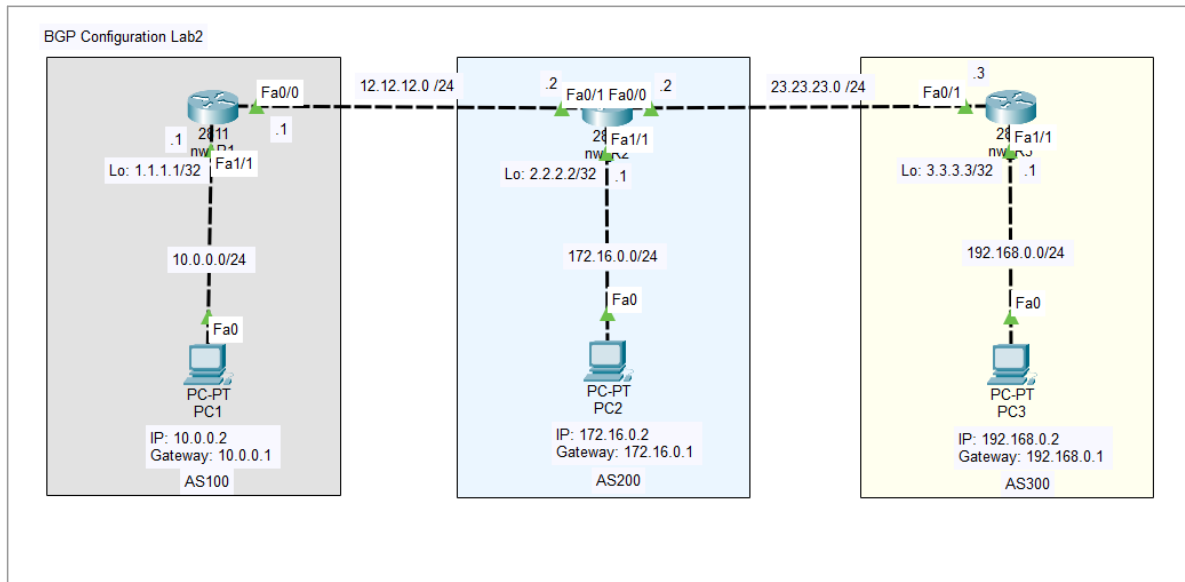


Figure 1 - Topologie BGP Configuration Lab2

Table d'adressage

Appareil	Interface	Adresse IP	Masque
R1 (AS 100)	Loopback0	1.1.1.1	255.255.255.255
R1	FastEthernet0/0 (vers R2)	12.12.12.1	255.255.255.0
R1	FastEthernet1/1 (LAN)	10.0.0.1	255.255.255.0
R2 (AS 200)	Loopback0	2.2.2.2	255.255.255.255
R2	FastEthernet0/1 (vers R1)	12.12.12.2	255.255.255.0
R2	FastEthernet0/0 (vers R3)	23.23.23.2	255.255.255.0
R2	FastEthernet1/1 (LAN)	172.16.0.1	255.255.255.0
R3 (AS 300)	Loopback0	3.3.3.3	255.255.255.255
R3	FastEthernet0/1 (vers R2)	23.23.23.3	255.255.255.0
R3	FastEthernet1/1 (LAN)	192.168.0.1	255.255.255.0
PC1	FastEthernet0	10.0.0.2	255.255.255.0 (GW 10.0.0.1)
PC2	FastEthernet0	172.16.0.2	255.255.255.0 (GW 172.16.0.1)
PC3	FastEthernet0	192.168.0.2	255.255.255.0 (GW 192.168.0.1)

2. Routeur nw_R1 (AS 100)

Configuration des interfaces (rappel)

```
Router> enable
Router# configure terminal
Router(config)# hostname nw_R1
nw_R1(config)# interface Loopback0
nw_R1(config-if)# ip address 1.1.1.1 255.255.255.255
nw_R1(config-if)# interface FastEthernet0/0
nw_R1(config-if)# ip address 12.12.12.1 255.255.255.0
nw_R1(config-if)# no shutdown
nw_R1(config-if)# interface FastEthernet1/1
nw_R1(config-if)# ip address 10.0.0.1 255.255.255.0
nw_R1(config-if)# no shutdown
nw_R1(config-if)# exit
```

Configuration BGP

Le voisin eBGP est R2 (12.12.12.2, AS 200). On annonce la loopback, le LAN 10.0.0.0/24 et le réseau de liaison 12.12.12.0/24.

```
nw_R1# configure terminal
nw_R1(config)# router bgp 100
nw_R1(config-router)# bgp router-id 1.1.1.1
nw_R1(config-router)# neighbor 12.12.12.2 remote-as 200
nw_R1(config-router)# network 1.1.1.1 mask 255.255.255.255
nw_R1(config-router)# network 10.0.0.0 mask 255.255.255.0
nw_R1(config-router)# network 12.12.12.0 mask 255.255.255.0
nw_R1(config-router)# end
nw_R1# copy running-config startup-config
```

```
nw_R1>en
nw_R1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
nw_R1(config)#router bgp 100
nw_R1(config-router)#bgp router-id 1.1.1.1
nw_R1(config-router)#neighbor 12.12.12.2 remote-as 200
nw_R1(config-router)#network 1.1.1.1 mask 255.255.255.255
nw_R1(config-router)#network 10.0.0.0 mask 255.255.255.0
nw_R1(config-router)#network 12.12.12.0 mask 255.255.255.0
nw_R1(config-router)#
```

Figure 2 - nw_R1 : processus BGP 100, router-id, voisin R2 et réseaux annoncés

3. Routeur nw_R2 (AS 200)

Configuration des interfaces (rappel)

```
Router> enable
Router# configure terminal
Router(config)# hostname nw_R2
nw_R2(config)# interface Loopback0
nw_R2(config-if)# ip address 2.2.2.2 255.255.255.255
nw_R2(config-if)# interface FastEthernet0/1
nw_R2(config-if)# ip address 12.12.12.2 255.255.255.0
nw_R2(config-if)# no shutdown
nw_R2(config-if)# interface FastEthernet0/0
nw_R2(config-if)# ip address 23.23.23.2 255.255.255.0
nw_R2(config-if)# no shutdown
nw_R2(config-if)# interface FastEthernet1/1
nw_R2(config-if)# ip address 172.16.0.1 255.255.255.0
nw_R2(config-if)# no shutdown
nw_R2(config-if)# exit
```

Configuration BGP

R2 a deux voisins eBGP : R1 (AS 100) et R3 (AS 300). Le message BGP-5-ADJCHANGE ... Up confirme l'établissement de la session avec R1.

```
nw_R2# configure terminal
nw_R2(config)# router bgp 200
nw_R2(config-router)# bgp router-id 2.2.2.2
nw_R2(config-router)# neighbor 12.12.12.1 remote-as 100
nw_R2(config-router)# neighbor 23.23.23.3 remote-as 300
nw_R2(config-router)# network 2.2.2.2 mask 255.255.255.255
nw_R2(config-router)# network 172.16.0.0 mask 255.255.255.0
nw_R2(config-router)# network 12.12.12.0 mask 255.255.255.0
nw_R2(config-router)# network 23.23.23.0 mask 255.255.255.0
nw_R2(config-router)# exit
nw_R2(config)# end
nw_R2# copy running-config startup-config
```

```
nw_R2>en
nw_R2#conf t
Enter configuration commands, one per line. End with CNTL/Z.
nw_R2(config)#router bgp 200
nw_R2(config-router)#bgp router-id 2.2.2.2
nw_R2(config-router)#neighbor 12.12.12.1 remote-as 100
nw_R2(config-router)##BGP-5-ADJCHANGE: neighbor 12.12.12.1 Up

nw_R2(config-router)#neighbor 23.23.23.3 remote-as 300
nw_R2(config-router)#network 2.2.2.2 mask 255.255.255.255
nw_R2(config-router)#network 172.16.0.0 mask 255.255.255.0
nw_R2(config-router)#network 12.12.12.0 mask 255.255.255.0
nw_R2(config-router)#network 23.23.23.0 mask 255.255.255.0
nw_R2(config-router)#exit
nw_R2(config)#end
nw_R2#
%SYS-5-CONFIG_I: Configured from console by console

nw_R2#en
nw_R2#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
nw_R2#
```

Figure 3 - nw_R2 : processus BGP 200, voisins R1 et R3, réseaux annoncés et sauvegarde

Vérification sur R2

```
nw_R2# show ip route
```

```
nw_R2#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is not set

  1.0.0.0/32 is subnetted, 1 subnets
B       1.1.1.1 [20/0] via 12.12.12.1, 00:00:00
  2.0.0.0/32 is subnetted, 1 subnets
C       2.2.2.2 is directly connected, Loopback0
 10.0.0.0/24 is subnetted, 1 subnets
B       10.0.0.0 [20/0] via 12.12.12.1, 00:00:00
 12.0.0.0/24 is subnetted, 1 subnets
C       12.12.12.0 is directly connected, FastEthernet0/1
 23.0.0.0/24 is subnetted, 1 subnets
C       23.23.23.0 is directly connected, FastEthernet0/0
 172.16.0.0/24 is subnetted, 1 subnets
C       172.16.0.0 is directly connected, FastEthernet1/1
--More--
```

Figure 4 - show ip route sur R2 : les routes B (1.1.1.1/32 et 10.0.0.0/24) sont apprises de R1 via 12.12.12.1 (eBGP, distance administrative 20)

4. Routeur nw_R3 (AS 300)

Configuration des interfaces (rappel)

```
Router> enable
Router# configure terminal
Router(config)# hostname nw_R3
nw_R3(config)# interface Loopback0
nw_R3(config-if)# ip address 3.3.3.3 255.255.255.255
nw_R3(config-if)# interface FastEthernet0/1
nw_R3(config-if)# ip address 23.23.23.3 255.255.255.0
nw_R3(config-if)# no shutdown
nw_R3(config-if)# interface FastEthernet1/1
nw_R3(config-if)# ip address 192.168.0.1 255.255.255.0
nw_R3(config-if)# no shutdown
nw_R3(config-if)# exit
```

Configuration BGP

```
nw_R3# configure terminal
nw_R3(config)# router bgp 300
nw_R3(config-router)# neighbor 23.23.23.2 remote-as 200
nw_R3(config-router)# network 3.3.3.3 mask 255.255.255.255
nw_R3(config-router)# network 192.168.0.0 mask 255.255.255.0
nw_R3(config-router)# network 23.23.23.0 mask 255.255.255.0
nw_R3(config-router)# end
nw_R3# copy running-config startup-config
```

```
nw_R3>en
nw_R3#conf t
Enter configuration commands, one per line. End with CNTL/Z.
nw_R3(config)#router bgp 300
nw_R3(config-router)#neighbor 23.23.23.2 remote-as 200
nw_R3(config-router)%%BGP-5-ADJCHANGE: neighbor 23.23.23.2 Up

nw_R3(config-router)#network 3.3.3.3 mask 255.255.255.255
nw_R3(config-router)#network 192.168.0.0 mask 255.255.255.0
nw_R3(config-router)#network 23.23.23.0 mask 255.255.255.0
nw_R3(config-router)#exit
nw_R3(config)#end
nw_R3#
%SYS-5-CONFIG_I: Configured from console by console

nw_R3#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
nw_R3#
```

Figure 5 - nw_R3 : processus BGP 300, voisin R2 (session Up), réseaux annoncés et sauvegarde

Vérification sur R3

```
nw_R3# show ip route
```

```
nw_R3#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

 1.0.0.0/32 is subnetted, 1 subnets
B    1.1.1.1 [20/0] via 23.23.23.2, 00:00:00
 2.0.0.0/32 is subnetted, 1 subnets
B    2.2.2.2 [20/0] via 23.23.23.2, 00:00:00
 3.0.0.0/32 is subnetted, 1 subnets
C    3.3.3.3 is directly connected, Loopback0
10.0.0.0/24 is subnetted, 1 subnets
B   10.0.0.0 [20/0] via 23.23.23.2, 00:00:00
12.0.0.0/24 is subnetted, 1 subnets
B   12.12.12.0 [20/0] via 23.23.23.2, 00:00:00
23.0.0.0/24 is subnetted, 1 subnets
C   23.23.23.0 is directly connected, FastEthernet0/1
--More--
```

Figure 6 - show ip route sur R3 : R3 apprend 1.1.1.1/32, 2.2.2.2/32, 10.0.0.0/24 et 12.12.12.0/24 via 23.23.23.2 (R2)

5. Tests de connectivité depuis les PC

Si les trois AS échangent correctement leurs routes, chaque PC joint les deux autres. Le TTL indique le nombre de routeurs traversés : 126 pour deux routeurs, 125 pour trois.

PC1 (10.0.0.2)

```
C:\> ping 172.16.0.2
C:\> ping 192.168.0.2
```

```
C:\>ping 172.16.0.2

Pinging 172.16.0.2 with 32 bytes of data:

Reply from 172.16.0.2: bytes=32 time=1ms TTL=126
Reply from 172.16.0.2: bytes=32 time<1ms TTL=126
Reply from 172.16.0.2: bytes=32 time<1ms TTL=126
Reply from 172.16.0.2: bytes=32 time<1ms TTL=126

Ping statistics for 172.16.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Figure 7 - PC1 vers PC2 : 4 réponses sur 4, TTL 126 (R1 puis R2)

```
C:\>ping 192.168.0.2

Pinging 192.168.0.2 with 32 bytes of data:

Reply from 192.168.0.2: bytes=32 time<1ms TTL=125
Reply from 192.168.0.2: bytes=32 time=1ms TTL=125
Reply from 192.168.0.2: bytes=32 time<1ms TTL=125
Reply from 192.168.0.2: bytes=32 time<1ms TTL=125

Ping statistics for 192.168.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

Figure 8 - PC1 vers PC3 : 4 réponses sur 4, TTL 125 (R1, R2 puis R3)

PC2 (172.16.0.2)

```
C:\> ping 10.0.0.2
C:\> ping 192.168.0.2
```

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.0.0.2

Pinging 10.0.0.2 with 32 bytes of data:

Reply from 10.0.0.2: bytes=32 time<1ms TTL=126
Reply from 10.0.0.2: bytes=32 time<1ms TTL=126
Reply from 10.0.0.2: bytes=32 time<1ms TTL=126
Reply from 10.0.0.2: bytes=32 time<1ms TTL=126

Ping statistics for 10.0.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.0.2

Pinging 192.168.0.2 with 32 bytes of data:

Reply from 192.168.0.2: bytes=32 time<1ms TTL=126
Reply from 192.168.0.2: bytes=32 time<1ms TTL=126
Reply from 192.168.0.2: bytes=32 time<1ms TTL=126
Reply from 192.168.0.2: bytes=32 time<1ms TTL=126

Ping statistics for 192.168.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Figure 9 - PC2 vers PC1 et PC3 : 0 % de perte, TTL 126 (un seul routeur de chaque côté de R2)

PC3 (192.168.0.2)

```
C:\> ping 10.0.0.2
C:\> ping 172.16.0.2
```

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.0.0.2

Pinging 10.0.0.2 with 32 bytes of data:

Reply from 10.0.0.2: bytes=32 time<1ms TTL=125
Reply from 10.0.0.2: bytes=32 time<1ms TTL=125
Reply from 10.0.0.2: bytes=32 time=4ms TTL=125
Reply from 10.0.0.2: bytes=32 time<1ms TTL=125

Ping statistics for 10.0.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 4ms, Average = 1ms

C:\>ping 172.16.0.2

Pinging 172.16.0.2 with 32 bytes of data:

Reply from 172.16.0.2: bytes=32 time<1ms TTL=126
Reply from 172.16.0.2: bytes=32 time<1ms TTL=126
Reply from 172.16.0.2: bytes=32 time<1ms TTL=126
Reply from 172.16.0.2: bytes=32 time<1ms TTL=126

Ping statistics for 172.16.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>|
```

Figure 10 - PC3 vers PC1 (TTL 125) et PC2 (TTL 126) : 0 % de perte

6. Récapitulatif

Commande	Équipement	Ce qu'elle montre
router bgp [AS]	R1, R2, R3	Active BGP avec le numéro d'AS local
neighbor [IP] remote-as [AS]	R1, R2, R3	Déclare le voisin eBGP et son AS
network [réseau] mask [masque]	R1, R2, R3	Annonce un préfixe dans BGP
show ip route	R2, R3	Routes B apprises par eBGP
ping	PC1, PC2, PC3	Connectivité de bout en bout entre les trois LAN